BBN and X-ray Viability of keV Sterile-Neutrino Dark Matter: A Tool-Assisted 2024–2026 Survey

HEP-Genesis-Agent session record
Project: ramp-research-sterile-bbn-xray

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Abstract

We report a tool-assisted study of which regions of the sterile-neutrino dark-matter plane $(m_s, \sin^2 2\theta)$ survive the combination of Big Bang nucleosynthesis (BBN) and X-ray decay-line constraints. The literature component draws mainly on 2024–2026 arXiv work. BBN observables (Y_p, D/H, $\Delta N_{\rm eff}$) came from a sterile-neutrino server that uses a fit to the PRyMordial network and takes $\Delta N_{\rm eff}$ as its only BBN input.

A 100×100 Dodelson–Widrow (DW) scan over $m_s = 1$–100 keV and $\sin^2 2\theta = 10^{-15}$–$10^{-6}$ shows that BBN ($\Delta\chi^2 > 3.84$) excludes only large mixings, $\sin^2 2\theta \gtrsim 10^{-8}$–$3.5 \times 10^{-7}$. Along the server's $\Omega_s h^2 = 0.12$ line, $\Delta N_{\rm eff} = 10^{-4}$–$1.1 \times 10^{-2}$ and $|\Delta\chi^2_{\rm BBN}| < 0.15$. The overclosure bound is therefore 40–5000 times stronger than BBN, and BBN is not a useful constraint on keV dark matter in the absence of a lepton asymmetry. The server's radiative decay rate agrees with the standard formula to about 1.5%.

Recent X-ray results provide the upper bounds on mixing. From the XRISM ten-cluster stack (arXiv:2510.24560) we derive $\sin^2 2\theta \lesssim 4 \times 10^{-10}$ at 7.1 keV. NuSTAR limits, as quoted secondarily in arXiv:2507.18752, give $\lesssim 10^{-12}$ at 6–40 keV, and INTEGRAL/SPI limits from the same source give $\lesssim 3 \times 10^{-14}$ above 40 keV.

Non-resonant DW production is disfavoured as all of the dark matter. The verdict at 7.1 keV depends on the relic normalization, and the server's normalization differs from a published fit (arXiv:1705.01837) by a factor of 10–17 in mixing. The surviving region quoted in the literature, Shi–Fuller production with $L \gtrsim 0.5$ at $m_s \gtrsim 10$ keV and $\sin^2 2\theta \lesssim 10^{-14}$, could not be tested with the available server. The synthesis table and several literature-derived limits remained contested by the session's critic.

1 Introduction

The user asked for a survey of recent (2024–2026) arXiv work on sterile-neutrino dark matter, and for BBN constraints to be combined with X-ray constraints to determine which $(m_s, \sin^2 2\theta)$ regions remain viable. BBN numbers (Y_p, D/H) were to come from the sterile-neutrino tools, and the answer was to be quantified with tool calls where possible.

The session organised this request into six lines of investigation:

1. BBN and relic-density constraints across the plane, from the sterile-neutrino server.

2. X-ray decay-line constraints from 2024–2026 limits, compared with server-computed line energies and rates.

3. A survey of 2024–2026 production mechanisms (Shi–Fuller resonant production with lepton asymmetry, reheating production, scalar decays), together with structure-formation bounds.

4. A synthesis into per-mechanism viability regions.

5. Re-extraction of contested literature numbers as verbatim quotes, with a limit table at $m_s = 7.1$, 10, 20, 40 and 50 keV.

6. Resolution of a factor-3 discrepancy in the DW relic normalization, the thermal-equivalent mass $m_{\rm th}$ along the DW line, and a BBN robustness check that includes an electron-neutrino degeneracy ξ_e.

The scope is limited to $m_s \lesssim 100$ keV, the server's validity range. The session did not run any resonant-production Boltzmann calculation, any BBN calculation with ξ_e, or any analysis of X-ray data.

2 Methods

All computations ran locally; the dispatch reported local execution. Three tool families were used.

Literature tools. The literature servers searched arXiv abstracts, retrieved metadata, downloaded full texts, read text chunks, extracted figures and recorded citations. Full texts were read for arXiv:2507.18752, arXiv:2405.17861, arXiv:2510.24560, arXiv:1705.01837 and arXiv:1702.08430. The text of astro-ph/0501562 was downloaded, but the mapping formula attributed to it was not quoted verbatim from it. Metadata only was obtained for arXiv:2502.17374, arXiv:2601.03346 and arXiv:2609.31340.

Sterile-neutrino server (`serve-sbbn.sh`). Its self-description gives the following model and data choices.

- BBN abundances (Y_p, D/H, ^{7}Li) come from a fit to the PRyMordial network, accurate to better than 0.3%. The only BBN input is $\Delta N_{\rm eff}$.
- The fit is valid for $N_{\rm eff} \in [2.044, 6.044]$, $\omega_b h^2 \in [0.019, 0.0257]$, $\tau_n \in [866.4, 890.4]$ s, and $m_s \lesssim 100$ keV. Decay injection is not modelled for $m_s \geq 1$ MeV.
- The observational inputs are $Y_p = 0.2449 \pm 0.004$ (Aver, Olive & Skillman 2021), D/H $= (2.547 \pm 0.025) \times 10^{-5}$ and Planck $N_{\rm eff} = 2.99 \pm 0.17$.
- A point is excluded if $\Delta\chi^2_{\rm BBN} > 3.84$ relative to the Standard Model (SM).
- Two mechanisms are available. In DW (non-resonant) mode, $\Delta N_{\rm eff}$ follows from the DW relic density. In Shi–Fuller (SF) mode, $\Delta N_{\rm eff}$ is a heuristic equal to $0.1\times$ the DW value, the relic density is a fitting formula in the lepton asymmetry L, L is capped at 0.1, and L is not fed into the BBN network.

The following server runs were made:

- a DW grid, $m_s = 7.1, 10, 20, 50$ keV $\times$ $\sin^2 2\theta = 10^{-10}, 10^{-12}, 10^{-14}$;
- a DW grid of 9 masses (1–100 keV) $\times$ 12 mixings (10^{-12}–3×10^{-7});
- zipped DW grids located on the $\Omega_s h^2 = 0.12$ line;
- a 100×100 DW scan, $m_s = 1$–100 keV and $\sin^2 2\theta = 10^{-15}$–$10^{-6}$, with its exclusion plot;
- SF grids at $m_s = 10, 20, 30$ keV with $L = 10^{-3}$ and 0.1;
- the BBN-versus-$N_{\rm eff}$ impact plot;
- a BEST short-baseline cross-check;
- X-ray line predictions at (7.1 keV, 10^{-10}), (20 keV, 10^{-12}), (10 keV, 10^{-12}) and (50 keV, 3×10^{-14}). X-ray fluxes are whole-halo with $f_{\rm DM} = 1$ and no field-of-view cut.

Hand calculations. Three analyses were done by hand rather than by tools.

- Inversion of the published DW fit.
- The DW-to-thermal mapping $m_s = 4.43\,\text{keV}\,(m_{\rm th}/\text{keV})^{4/3}$.
- Linearised BBN χ^2 bounds, using the server slopes $\mathrm{d}Y_p/\mathrm{d}N_{\rm eff} = 0.0135$ and $\mathrm{d}(\mathrm{D/H} \times 10^5)/\mathrm{d}N_{\rm eff} = 0.331$ about the server SM values. The joint fit used $\mathrm{d}Y_p/\mathrm{d}\xi_e = -0.25$.

Research board and supplement. Findings, dead ends and critic verdicts were tracked on a research board. The full tool-call log and the code are in the supplement.

3 Results

3.1 Literature survey

Four 2025–2026 papers on production mechanisms were confirmed to exist by metadata lookup.

- Vogel, García Escudero, Froustey and Abazajian, arXiv:2507.18752. This paper revives Shi–Fuller production.
- Gorbunov, Kalashnikov and Krugan, arXiv:2502.17374. This paper studies flavor-dependent lepton asymmetry.
- Cline and Xu, arXiv:2601.03346. This paper studies cold sterile-neutrino dark matter from inflaton decays during reheating, with branching ratio BR $\lesssim 10^{-4}$.
- Gorbunov and Kalashnikov, arXiv:2609.31340. This paper studies saturation of resonant production at high flavor asymmetry.

The full text of arXiv:2507.18752 was read. It states that "lepton asymmetries $L \gtrsim 0.5$ at high temperatures open significant viable parameter space in the $m_s \gtrsim 10$ keV and $\sin^2(2\theta) \lesssim 10^{-14}$ regime." It cites the following limits:

- NuSTAR: $\sin^2(2\theta) \lesssim 10^{-12}$ in the 6–40 keV mass range.
- INTEGRAL/SPI: "exclude lifetimes below $\sim 10^{20}$ s at 95% CL [corresponding to $\sin^2(2\theta) \lesssim 3 \times 10^{-14}$] in the $\gtrsim 40$ keV mass regime."

- Structure formation: $m_{\rm th} > 8.4$ keV (galaxy counts) and $m_{\rm th} > 8.5$ keV (counts plus strong lensing), both at 95% CL, and a Lyman-α preference $m_{\rm th} = 4.1^{+33.3}_{-1.5}$ keV.

The paper also states that the original DW mechanism is ruled out as producing all of the dark matter.

The claims of arXiv:2502.17374, arXiv:2601.03346 and arXiv:2609.31340 are known only from their abstracts.

- arXiv:2502.17374: flavor asymmetries that cancel in active oscillations, more efficiently for inverted ordering, mitigate BBN limits. Asymmetry that disappears mid-production yields colder spectra that reopen lighter masses.
- arXiv:2601.03346: reheating production reaches mixings small enough to evade X-ray limits.
- arXiv:2609.31340: oscillation-induced redistribution saturates the abundance at large initial asymmetry.

On the X-ray side, two recent papers were found and read: the XRISM Resolve ten-cluster stack (arXiv:2510.24560) and the 11-year NuSTAR stray-light analysis (arXiv:2405.17861). Merle's 2017 review (arXiv:1702.08430) served only as older context.

A web snippet mentioned eROSITA eFEDS line limits over 1.8–18 keV. No citable arXiv identifier was found for it, so it was not used. In its initial survey summary, written before the server was used, the assistant cautioned that it had no BBN tool and that its BBN statements were abstract-level. That gap was closed in the subsequent waves.

3.2 BBN versus the DW plane

The DW scan of 10^4 points flagged 1338 points as BBN-excluded and 4300 as overclosing ($\Omega_s h^2 > 0.12$). The largest BBN-allowed mixing at each mass is listed in Table 1, alongside the server's dark-matter line.

m_s (keV)	BBN-allowed max $\sin^2 2\theta$	DW line ($\Omega_s h^2 = 0.12$, server)
1	3.5×10^{-7}	8.7×10^{-9}
3–3.7	1.2×10^{-7} (3.7 keV)	1.2×10^{-9} (3 keV)
7–7.1	8.1×10^{-8}	2.55×10^{-10}
10–13.5	4.3×10^{-8} (13.5 keV)	1.37×10^{-10} (10 keV)
20	–	3.9×10^{-11}
50	1.5×10^{-8}	7.6×10^{-12}
100	1×10^{-8}	2.2×10^{-12}

Table 1: BBN-allowed upper mixing from the DW scan, and the server's DW dark-matter line. The line was also computed at 2 keV (2.5×10^{-9}) and 5 keV (4.8×10^{-10}).

Along the dark-matter line, $\Delta N_{\rm eff} = 10^{-4}$–$1.1 \times 10^{-2}$ and $|\Delta\chi^2_{\rm BBN}| < 0.15$. Overclosure is thus 40 times (1 keV) to about 5000 times (100 keV) more restrictive than BBN. The entire DW line, and everything below it, is BBN-allowed.

The benchmark grid at 7.1–50 keV and 10^{-14}–10^{-10} is consistent with this: no point was excluded, $\Delta N_{\rm eff} \leq 2.9 \times 10^{-3}$, and $|\Delta\chi^2| \leq 0.04$. The same grid gives the DW abundance at 10^{-10}:

- $\Omega_s h^2 = 0.087$ at 10 keV;

- $\Omega_s h^2 = 0.30$ at 20 keV, which overcloses;
- $\Omega_s h^2 = 1.58$ at 50 keV, which overcloses.

At 10^{-12}–10^{-14} the DW abundance is $\Omega_s h^2 \leq 0.016$.

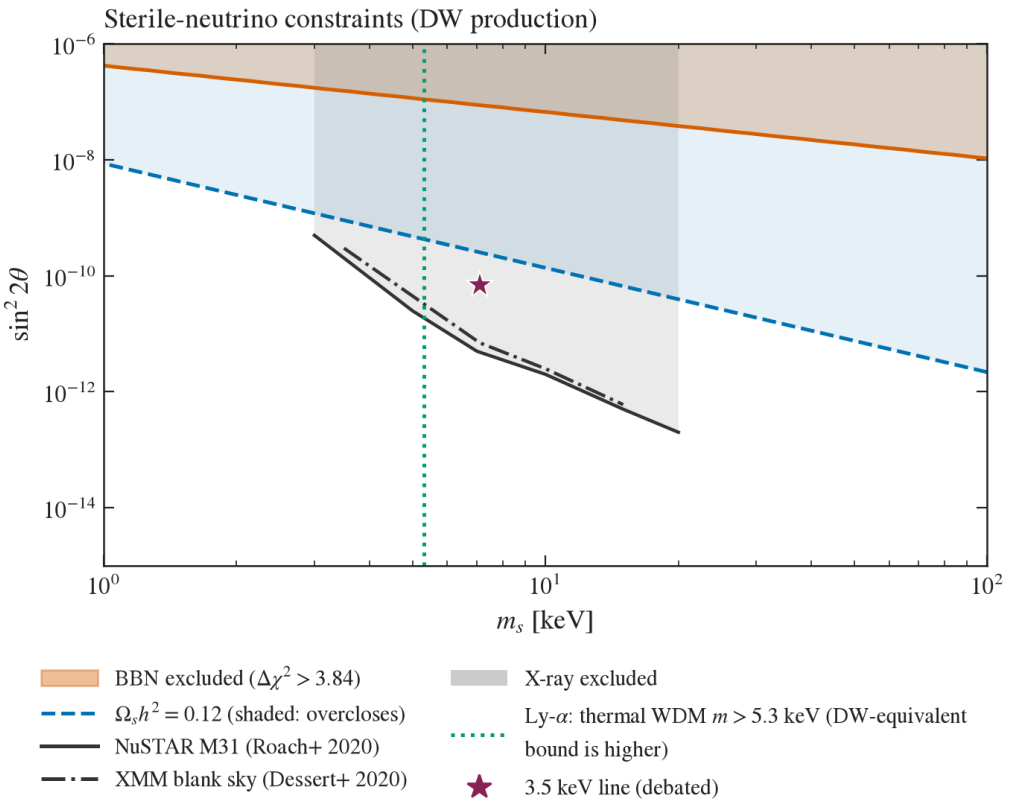


Figure 1: exclusion_scan_DW_dc7743.png Produced in this session by `plot_sterile_nu_exclusion` on *serve-sbbn.sh*.

Figure 1 shows the exclusion plane produced from this scan. The BBN-excluded band sits at large mixing, and the overclosure region lies well below it. This visualises the hierarchy in Table 1: for DW production the binding cosmological constraint is the abundance, not nucleosynthesis.

Representative BBN points. Table 2 lists server outputs at selected points, with deviations quoted against the brief's data ($Y_p = 0.245 \pm 0.003$ and D/H $= (2.53 \pm 0.03) \times 10^{-5}$).

Point	$\Delta N_{\rm eff}$	Y_p	D/H $\times 10^5$	$\Delta\chi^2$	Verdict
SM ($N_{\rm eff} = 3.044$)	0	0.246815 ($+0.6\sigma$)	2.4433 (-2.9σ)	0	baseline
7.1 keV, 3×10^{-10}	–	0.24684	2.4440	-0.024	allowed
1 keV, 3×10^{-7}	0.387	0.25193 ($+2.3\sigma$)	2.5718 ($+1.4\sigma$)	$+0.44$	allowed
10 keV, 10^{-7}	0.813	0.25730 ($+4.1\sigma$)	2.7145 ($+6.2\sigma$)	13.4	excluded
3 keV, 3×10^{-7}	0.931	0.25874 ($+4.6\sigma$)	2.7540 ($+7.5\sigma$)	19.3	excluded

Table 2: Server BBN outputs at representative DW points.

The 1 keV point is allowed despite its Y_p excess. The server's SM D/H lies below the data, so additional radiation improves the D/H fit and offsets the Y_p penalty.

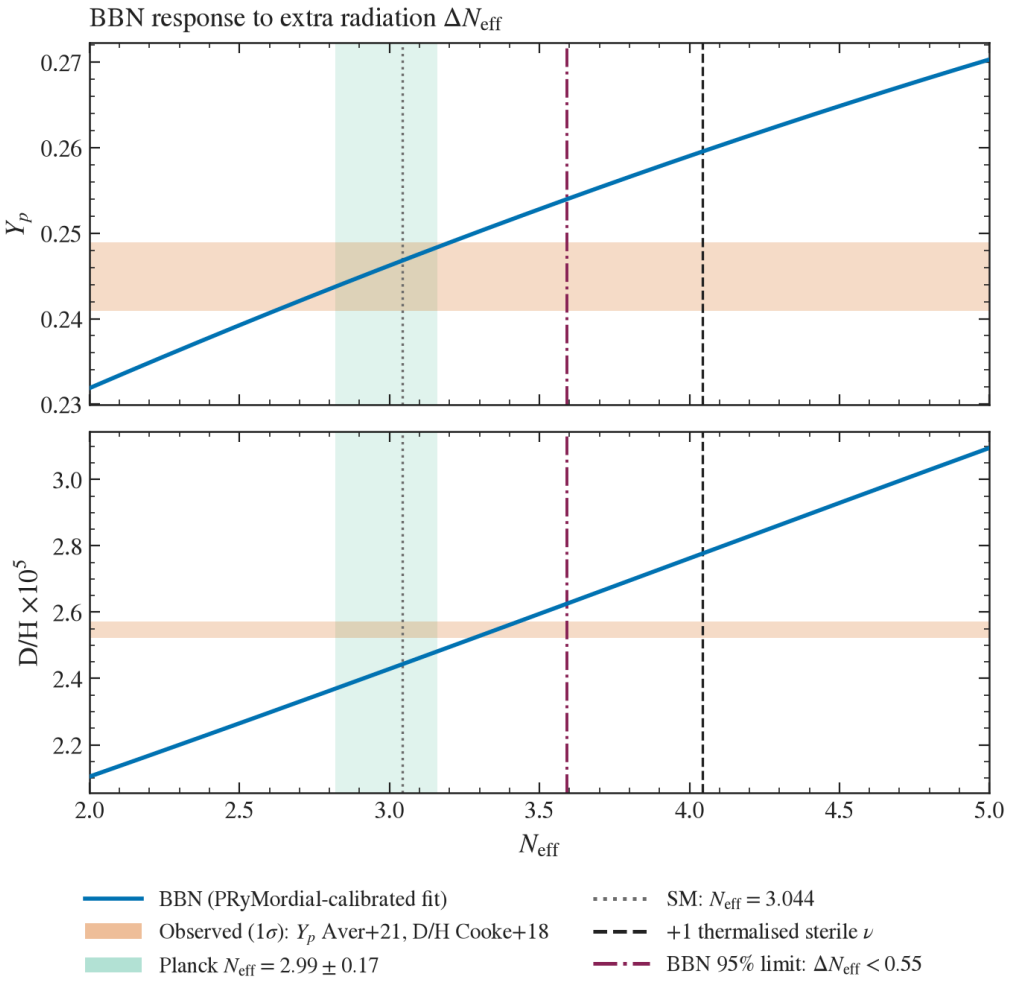


Figure 2: bbn_vs_neff.png Produced in this session by `plot_sterile_nu_bbn_impact` on *serve-sbbn.sh*.

Figure 2 shows the server's Y_p and D/H as functions of $N_{\rm eff}$, with the observational bands. A fully thermalised sterile state ($N_{\rm eff} = 4.044$) gives $Y_p = 0.2596$ and D/H $= 2.777 \times 10^{-5}$, which is 3.7σ above the observed Y_p. BBN alone limits $\Delta N_{\rm eff} < 0.548$ at 95%, with slopes $\mathrm{d}Y_p/\mathrm{d}N_{\rm eff} = 0.01351$ and $\mathrm{d}(\mathrm{D/H} \times 10^5)/\mathrm{d}N_{\rm eff} = 0.3308$.

The BEST cross-check ($m_s = 1.14$ eV, $\sin^2 2\theta = 0.42$) gave the following, although the critic had judged this check tangential to the keV question:

- $\Delta N_{\rm eff} = 1$, $Y_p = 0.25956$, $\Delta\chi^2 = 23.1$, so the point is excluded;
- 6.2σ tension with Planck, and a projected 33σ at CMB-S4;
- $\Omega_s h^2 = 29.3$.

3.3 BBN robustness: data baseline and ξ_e

These bounds come from a linearised hand analysis, not from server runs. Table 3 gives 95% upper limits on $\Delta N_{\rm eff}$ for $\Delta\chi^2 = 3.84$ relative to the SM; values in brackets are relative to the best fit.

The server's SM D/H lies 2.9σ (brief data) to 4.1σ (server data) below the observations. Because of this, D/H alone prefers $\Delta N_{\rm eff} \approx 0.26$–$0.31$, and the bound shifts by 0.1–0.2 depending

Data	Y_p only	D/H only	combined
Brief (0.245 ± 0.003; 2.53 ± 0.03)	0.32	0.58 [0.44]	0.47 [0.37]
Server (0.2449 ± 0.004; 2.547 ± 0.025)	0.46	0.66 [0.46]	0.61 [0.43]

Table 3: Linearised 95% bounds on $\Delta N_{\rm eff}$. The server's own reported limit is 0.548.

on the baseline.

The joint $(\Delta N_{\rm eff}, \xi_e)$ fit used $\mathrm{d}Y_p/\mathrm{d}\xi_e = -0.25$, assumed D/H independent of ξ_e, and applied two degrees of freedom ($\Delta\chi^2 = 5.99$).

- Brief data: $\Delta N_{\rm eff} \in [0.04, 0.48]$ with $\xi_e = 0.0073 + 0.054\,\Delta N_{\rm eff} \pm 0.029$.
- Server data: $\Delta N_{\rm eff} \in [0.13, 0.50]$ with $\xi_e = 0.0077 + 0.054\,\Delta N_{\rm eff} \pm 0.039$.

Overall this implies $|\xi_e| \lesssim 0.03$–0.06. A positive $\xi_e \approx 0.03$ compensates $\Delta N_{\rm eff} \approx 0.5$ in Y_p. Neglecting $\mathrm{d(D/H)}/\mathrm{d}\xi_e$ biases this contour.

3.4 Shi–Fuller runs with the server

SF runs at $m_s = 10, 20, 30$ keV with $\sin^2 2\theta = 10^{-14}$ gave $\Delta N_{\rm eff} \sim 10^{-8}$ and $\Delta\chi^2_{\rm BBN} = 0$. At 20 keV the outputs were $Y_p = 0.24682$ and $\mathrm{D/H} = 2.4433 \times 10^{-5}$. The relic abundance $\Omega_s h^2$ at the three masses was:

- $L = 10^{-3}$: 0.122, 0.490 and 1.10;
- $L = 0.1$: 1.22, 4.90 and 11.0.

All of these points were flagged as overclosing.

These runs do not test arXiv:2507.18752, for four reasons:

- L is capped at 0.1, while the paper requires $L \gtrsim 0.5$;
- the SF $\Delta N_{\rm eff}$ is a heuristic;
- ξ_e does not enter Y_p;
- the SF abundance is a fitting formula, not a Boltzmann calculation.

The BBN "allowed" verdict here is trivial, and the overclosure flags do not contradict the paper.

3.5 X-ray decay rate and limits

The server's rate agrees with $\Gamma = 1.38 \times 10^{-32}\,\mathrm{s}^{-1}\,(\sin^2 2\theta/10^{-10})(m_s/\mathrm{keV})^5$ to about 1.5%.

- At 7.1 keV and 10^{-10}: 2.46×10^{-28} s^{-1} (formula 2.49×10^{-28}), with $E_\gamma = 3.55$ keV and lifetime 4.07×10^{27} s.
- At 20 keV and 10^{-12}: 4.36×10^{-28} s^{-1} (formula 4.42×10^{-28}), with $E_\gamma = 10$ keV.

Eq. 2 of arXiv:2405.17861, $\Gamma_\gamma = 1.36 \times 10^{-22}(m_s/1\,\mathrm{keV})^5 \sin^2 2\theta$ s^{-1}, is consistent with this formula.

The whole-halo fluxes at (7.1 keV, 10^{-10}) are 4.68 ph s^{-1} cm^{-2} for the Milky Way centre, 6.89×10^{-4} for M31 and 3.55×10^{-5} for Perseus. The server's built-in limits, NuSTAR M31 (Roach+ 2020) and XMM blank sky (Dessert+ 2020), exclude both test points.

XRISM. The XRISM Resolve stack (arXiv:2510.24560) combines 3.75 Ms on ten clusters over 2.5–15 keV and detects no unidentified lines. Its 3σ limit at $m_s \sim 7.1$ keV is $\Gamma \sim 1.0 \times 10^{-27}$ s^{-1}. This is "3−4 times lower than" the Hitomi Perseus limit "but still 5 times higher than the XMM-Newton detection reported by Bulbul et al. (2014)," so the 3.55 keV line is constrained but not excluded. Scaling the server rate gives $\sin^2 2\theta \lesssim 4 \times 10^{-10}$; this conversion is the session's own, not a quoted value.

NuSTAR. arXiv:2405.17861 sets 95% limits for $m_s = 6$–40 keV. It states that the limits "reach the bottom edge of theoretical predictions," that observed limits are stronger than expected at 8–9 and 23–25 keV, and that there is an instrumental excess at 40 keV. Numerical limit values could not be read from the paper: Fig. 3 was absent from the HTML rendering, and the downloaded Fig. 1 image could not be digitised.

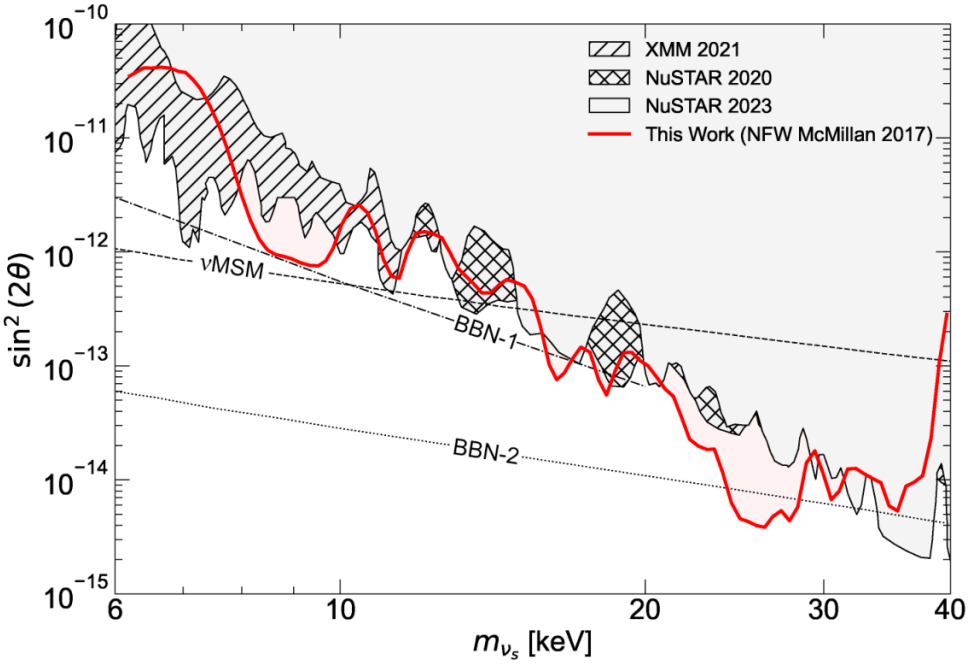


Figure 3: Figure 1: 95% upper limits on sin^{2}(2\theta) derived from the NuSTAR unfocused stray light spectrum for reference profile [15] along with some previous constraints [9, NuSTAR 2020, 2023:], [16, XMM 2021:] and updated theoretical lower bounds [17, 18, 19]. Note that the borderline excess at 40 keV is the known instrumental effect caused by absorbed stray light [20, 21, 22]. [arXiv:2405.17861] Reproduced from arXiv:2405.17861 (extracted with `get_paper_figures`).

Figure 3 reproduces Fig. 1 of arXiv:2405.17861. According to its caption, it shows the 95% upper limits on $\sin^2(2\theta)$ from the NuSTAR unfocused stray-light spectrum for the reference profile, together with earlier NuSTAR and XMM constraints and updated theoretical lower bounds. It attributes the borderline excess at 40 keV to absorbed stray light. No number in this report was read from the figure. It documents qualitatively that the 6–40 keV NuSTAR limits approach the theoretical lower bounds.

Limit table. The X-ray limit table assembled in the session is given in Table 4. It was contested by the critic, because only the XRISM entry derives from a primary source; the NuSTAR and SPI values are secondary quotes from arXiv:2507.18752.

m_s	Limit on $\sin^2 2\theta$	Server line-tool check
7.1 keV	XRISM $\lesssim 4 \times 10^{-10}$ (3σ, derived); NuSTAR $\lesssim 10^{-12}$	10^{-10} excluded by built-in tables
10 keV	NuSTAR $\lesssim 10^{-12}$	at 10^{-12}: $\Gamma = 1.36 \times 10^{-29}$ s^{-1}, $E = 5$ keV
20 keV	NuSTAR $\lesssim 10^{-12}$	10^{-12} excluded (built-in NuSTAR M31 tab
40 keV	NuSTAR upper edge; SPI $\lesssim 3 \times 10^{-14}$ above	–
50 keV	SPI $\lesssim 3 \times 10^{-14}$	at 3×10^{-14}: $\Gamma = 1.28 \times 10^{-27}$ s^{-1}, $E = 25$

Table 4: X-ray limits versus mass (95% CL unless noted). Contested.

The session flagged an inconsistency in the SPI entry. The quote equates "lifetimes below $\sim 10^{20}$ s" with $\sin^2 2\theta \lesssim 3 \times 10^{-14}$, but the server gives a lifetime of 7.84×10^{26} s (2.48×10^{19} yr) at 50 keV and that mixing. The assistant noted that the year-valued lifetime is close to 10^{20}; the record does not resolve the discrepancy further.

3.6 DW normalization and structure formation

The server's DW abundance scales as $\Omega_s h^2 = 0.1(\sin^2 2\theta/10^{-9})(m_s/3\,\mathrm{keV})^{1.8}$. This was checked at three points: 3 keV and 10^{-9} gives 0.100; 10 keV and 10^{-9} gives 0.873, a ratio of 8.73 that equals $(10/3)^{1.8}$; and 1 keV and 10^{-8} gives 0.138. The exponent matches the brief's formula, but the normalization corresponds to three times smaller mixing on the dark-matter line.

The published fit in the Abazajian review (arXiv:1705.01837, Eq. 37) is

$$m_s \sim 3.40\,\mathrm{keV}\,(\sin^2 2\theta/10^{-8})^{-0.615}(\Omega_{\mathrm{DM}}/0.26)^{0.5} \times 0.527\,\mathrm{erfc}[-1.15(T_{\mathrm{QCD}}/170\,\mathrm{MeV})^{2.15}].$$

Inverting it by hand gives the dark-matter line at $\sin^2 2\theta \sim 1.23 \times 10^{-8}$ (3 keV), 3.0×10^{-9} (7.1 keV), 1.73×10^{-9} (10 keV), 5.6×10^{-10} (20 keV) and 1.26×10^{-10} (50 keV). Varying T_{QCD} over 150–200 MeV shifts these by roughly $+10\%/-7\%$. Compared with the server line, the published fit requires 10–17 times more mixing; compared with the brief's formula, 3.4–5.5 times more.

At 7.1 keV the DW verdict against the XRISM-derived limit therefore depends on the normalization:

- the server line (2.55×10^{-10}) survives;
- the brief's line (7.65×10^{-10}) is excluded;
- the published line (3.0×10^{-9}) is excluded by about a factor of 7.5.

The assistant concluded that server-based DW statements are not conservative. This conclusion and the normalization comparison were both marked contested.

Thermal-equivalent mass. The mapping $m_s = 4.43\,\mathrm{keV}\,(m_{\mathrm{th}}/\mathrm{keV})^{4/3}$ was attributed to Viel et al. (astro-ph/0501562), but was not quoted verbatim from that paper. It gives $m_{\mathrm{th}} = 0.75$, 1.42, 1.84, 3.10 and 6.16 keV at $m_s = 3$, 7.1, 10, 20 and 50 keV. Lyman-α bounds of $m_{\mathrm{th}} > 4$ (6) keV then require $m_s > 28$ (48) keV, and satellite counts ($m_{\mathrm{th}} > 8.4$ keV) require $m_s > 76$ keV. At those masses the DW line, about 3×10^{-10} (published) or about 10^{-11} (server), lies far above the NuSTAR and SPI ceilings. On this basis DW is excluded by the combination of structure formation and X-ray limits regardless of normalization.

3.7 Per-mechanism viability

The final viability table (contested) reads as follows.

- **DW.** Not viable as all of the dark matter. The binding constraints are X-ray limits combined with structure formation; BBN is non-binding, with $\Delta\chi^2 < 0.15$ along the line.
- **Shi–Fuller with $L \gtrsim 0.5$ above 20 MeV.** Quoted as viable at $m_s \gtrsim 10$ keV and $\sin^2 2\theta \lesssim 10^{-14}$, below the NuSTAR and SPI ceilings. This rests on arXiv:2507.18752 and was not tool-verified.
- **Flavor-dependent asymmetry.** Lower mixings and lighter masses are reopened (arXiv:2502.17374, abstract only).
- **Large-L saturation.** This acts as an additional constraint on resonant production (arXiv:2609.31340, abstract only).
- **Reheating production.** Mixings below the X-ray limits are reached with BR $\lesssim 10^{-4}$ (arXiv:2601.03346, abstract only).

4 Discussion

The robust computational result is negative but important. For keV sterile neutrinos without lepton asymmetry, the extra radiation on and below the DW dark-matter line is $\Delta N_{\rm eff} \lesssim 10^{-2}$. Y_p and D/H then cannot be distinguished from standard BBN, and BBN carries essentially no constraining power where dark matter lives. This was the critic's expectation at the planning stage, and the scan confirmed it quantitatively. BBN becomes relevant to sterile-neutrino dark matter only through the lepton asymmetry needed for resonant production. That effect the server cannot model, because ξ_e is not an input.

The linearised robustness check shows that the BBN bound on $\Delta N_{\rm eff}$ shifts by 0.1–0.2 between data choices. It also shows that the low SM D/H of the server (2.9–4.1σ below the data) mildly favours extra radiation. The session raised, without resolving, whether this low D/H reflects a nuclear-rate choice that biases the server's bounds.

Several tensions limit the strength of the X-ray synthesis.

- The server's DW normalization disagrees with the published fit by an order of magnitude, so the 7.1 keV verdict flips between normalizations.
- The NuSTAR limits could not be extracted from the primary paper. The per-mass entries rest on a summary value of $\lesssim 10^{-12}$ quoted in arXiv:2507.18752.
- The server's own NuSTAR M31 table excludes 10^{-12} at 20 keV, which indicates that the summary value is not uniformly representative.
- The SPI lifetime–mixing correspondence is internally inconsistent with the computed lifetime.
- The DW-to-thermal mapping was not verified verbatim.

The critic left the following findings contested: the arXiv:2507.18752 extraction (later re-supported verbatim), the mechanism map, the XRISM conversion (later re-supported verbatim), the NuSTAR reading, the X-ray limit table, the final viability table, the normalization comparison and the 7.1 keV verdict. The critic's final verdict on the report was "revise."

Two literature searches were dead ends. Neither search tool returned a 2024–2026 eROSITA line-search paper, and a targeted search found no primary 2024+ INTEGRAL/SPI line limit at $m_s \geq 40$ keV. The eROSITA eFEDS statement therefore remained an uncited web snippet, and the SPI entry depends wholly on a secondary quote.

5 Conclusions

Established with tools:

- BBN excludes DW sterile neutrinos only above $\sin^2 2\theta \sim 10^{-8}$–$3.5 \times 10^{-7}$ over 1–100 keV. This is 40–5000 times weaker than overclosure, and the DW dark-matter line is BBN-allowed with $|\Delta\chi^2| < 0.15$.
- The server's BBN-only limit is $\Delta N_{\rm eff} < 0.548$; linearised alternatives span 0.32–0.66 depending on data and probe.
- The server's radiative rate is accurate to about 1.5%.

Supported by the literature but partially contested:

- DW production cannot be all of the dark matter once X-ray limits and structure formation are combined.
- The surviving region requires very small mixing, about $\lesssim 10^{-14}$ at $m_s \gtrsim 10$ keV for Shi–Fuller production with $L \gtrsim 0.5$, or non-oscillation production. Neither was tested numerically.

Open questions raised in the session:

- a Shi–Fuller BBN test requiring a code that accepts ξ_e (PRIMAT, AlterBBN, or PRyMordial with ξ);
- numerical NuSTAR stray-light limits at 7, 10 and 20 keV, and a 50 keV limit from SPI or the NuSTAR Galactic Centre analysis;
- whether the server's DW relic formula should be replaced by the Abazajian fit;
- whether the server's SM D/H biases its $\Delta N_{\rm eff}$ bounds.

References

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- arXiv:2609.31340 — D. Gorbunov, D. Kalashnikov, "Natural Saturation Of The Sterile Neutrino Dark Matter Resonant Production By a High Lepton Flavor Asymmetry In Primordial Plasma" (2026).
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- arXiv:2405.17861 — R. A. Krivonos, V. V. Barinov, A. A. Mukhin, D. S. Gorbunov, "Strong limits on keV-scale galactic sterile neutrino dark matter with stray light from NuSTAR after 11 years of operation" (2024).
- arXiv:1705.01837 — Abazajian, "Sterile neutrinos in cosmology."
- astro-ph/0501562 — Viel et al., "Constraining Warm Dark Matter candidates including sterile neutrinos and light gravitinos with WMAP and the Lyman-alpha forest" (2005).
- arXiv:1702.08430 — A. Merle, "keV sterile neutrino Dark Matter" (2017).

Data and code availability

All outputs of this session are collected in the project folder `/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray`. The session produced 14 data files; the supplement lists them. Full documentation of how these results were obtained is in the supplementary material (`supplement.pdf`: implementation details, the tool-call log, the research board with every finding and its evidence, figures not shown here, and the code index) and the verbatim transcript (`transcript.pdf`), generated alongside this report.